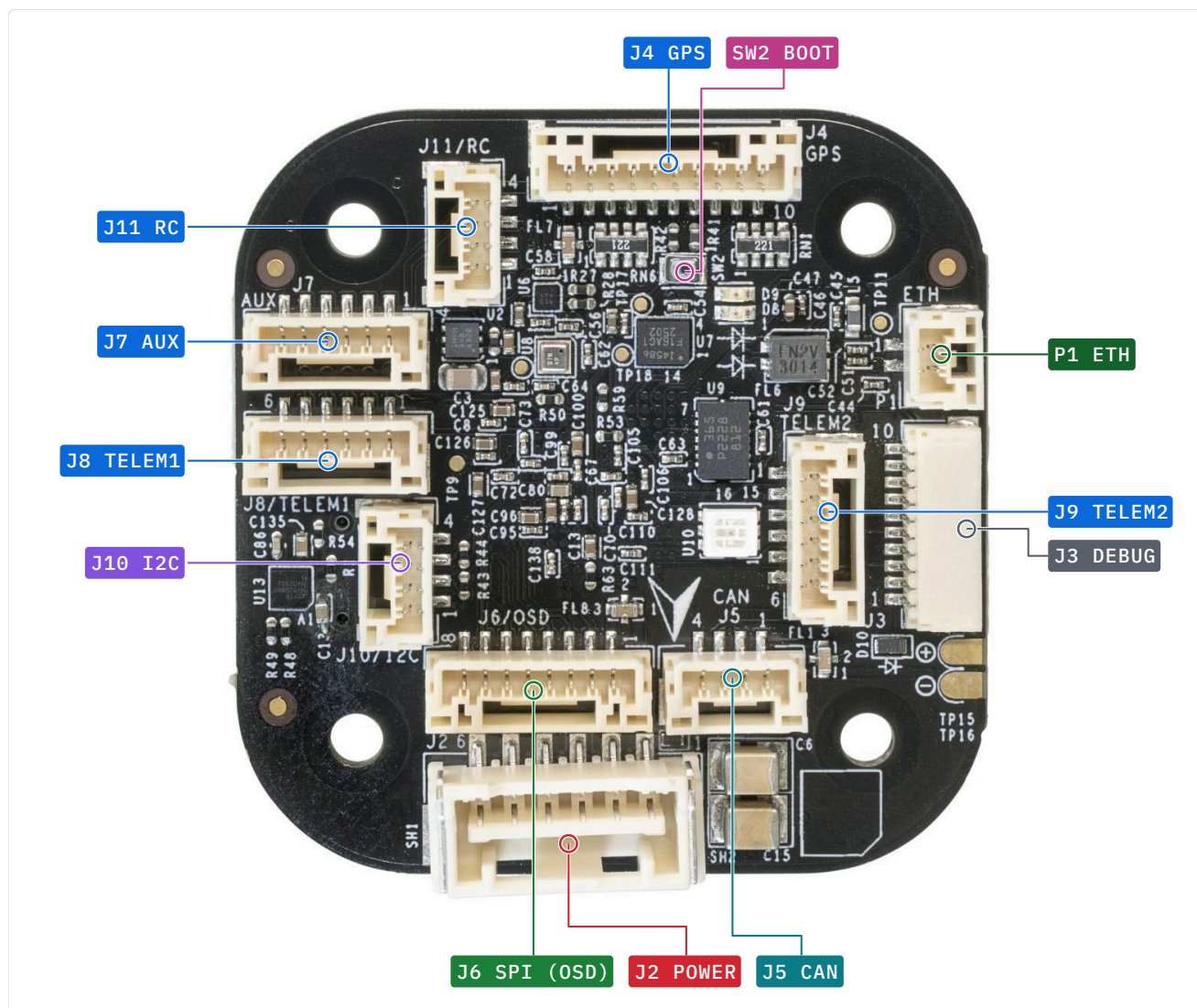


MR-VMU-Tropic hardware reference

Connectors and pinouts of the NXP MR-VMU-Tropic vehicle management unit, an open flight controller built around the i.MX RT1064. All data comes from the rev F schematic in the [MR-VMU-TROPIC repository: SPF-94468_F.pdf](#). The photos show a rev C board; the connector positions are the same.

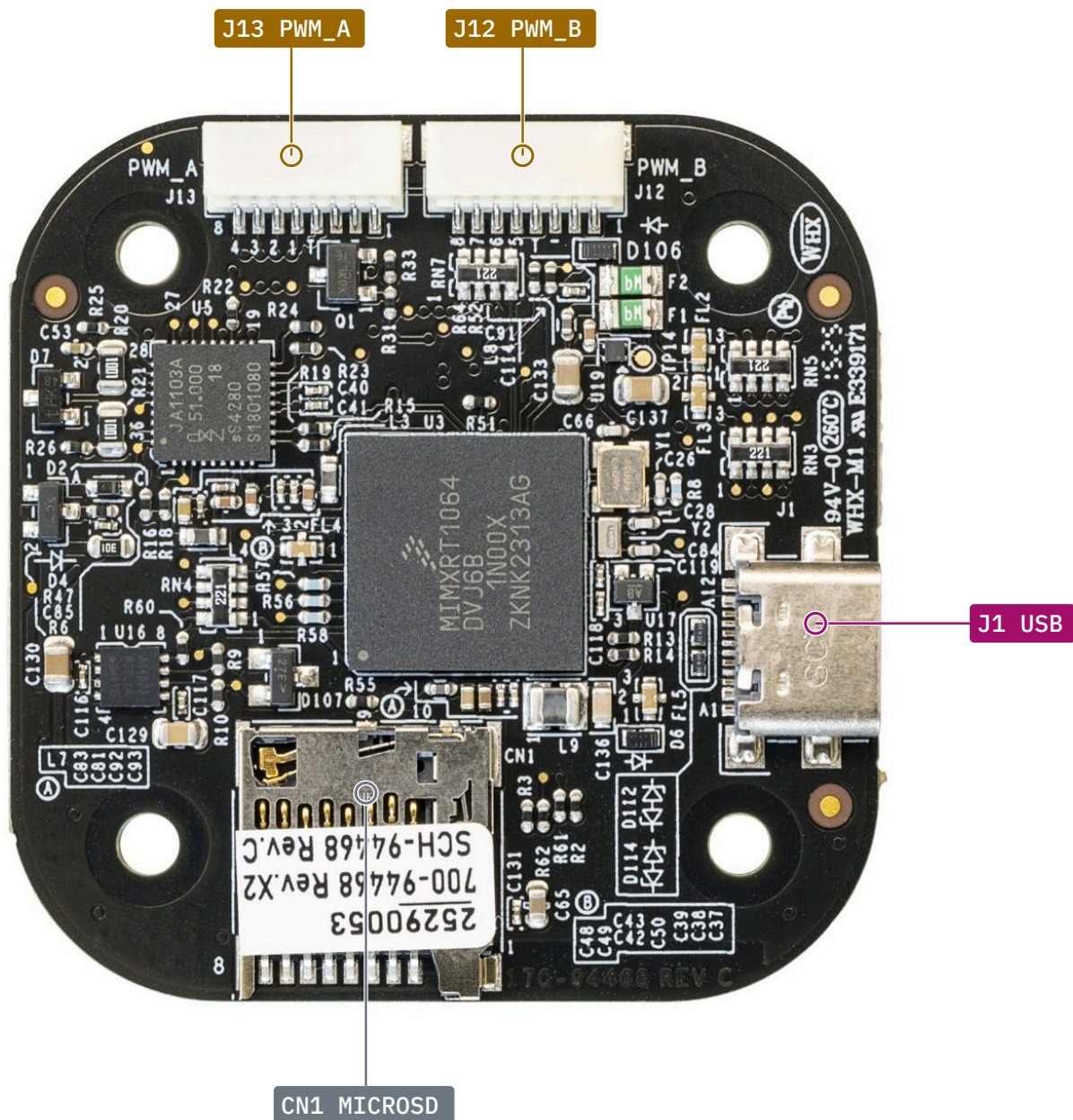
Each card shows three things per pin: the **function**, the **net name in the schematic** and the **i.MX RT1064 pad** the signal ends up on. Parts in between, such as the CAN transceiver and the Ethernet PHY, are shown too. Pin 1 is the square pad.

Board overview



Not visible here: J1 USB (bottom view) · J12 PWM_B (bottom view) · J13 PWM_A (bottom view) · CN1 MICROSD (bottom view)

Bottom side, with the two ESC headers, the USB-C port and the microSD slot:



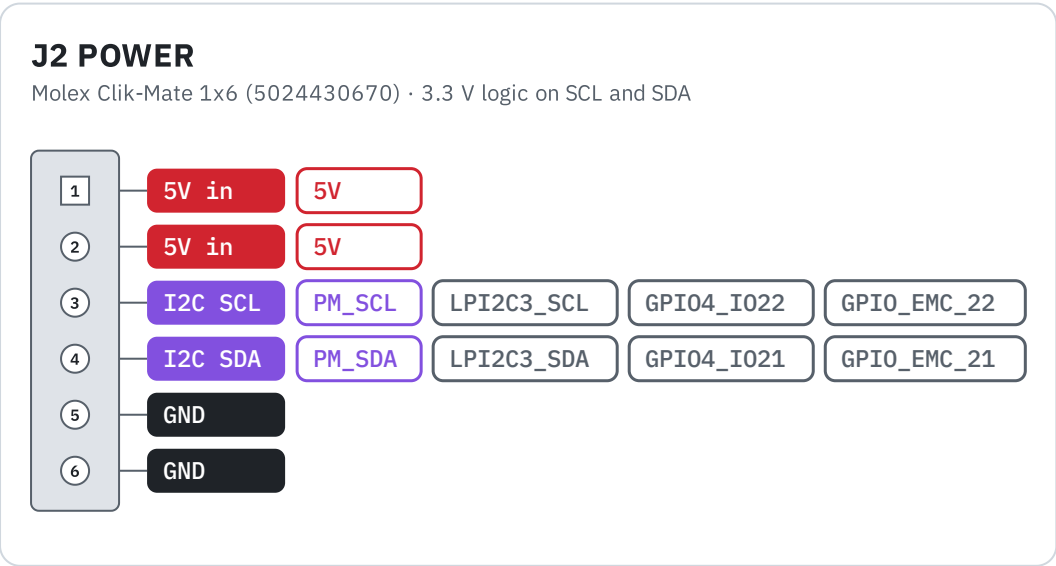
Connector index

Ref	Function	Connector	Details
Main CN1 MICROSD	microSD card slot	microSD push-push, 8 pins plus card detect	Four-bit SD interface on the first SD host of the chip, used for logging. The card detect switch goes to a separate chip pin.
Main J1 USB	USB 2.0 Type-C device port	USB Type-C receptacle (USB4110-GF-A)	Serial download port of the i.MX RT1064 boot ROM, and the port used for flashing. The board is always a USB device. VBUS can also power the board through protection switch U13.
Main P1 ETH	100BASE-T1 single pair Ethernet	JST-GH 1x2, top entry	
Main J2 POWER	Power module input, 5 V plus battery monitor I2C	Molex Clik-Mate 1x6 (5024430670)	

Ref	Function	Connector	Details
Main J3 DEBUG	SWD debug and console, DS-009 debug full layout	JST-SH 1x10	
Main J4 GPS	GPS module with compass, safety switch and buzzer	JST-GH 1x10 (BM10B-GHS-TBT), top entry	
Main J5 CAN	CAN FD (TJA1462)	JST-GH 1x4, top entry	
Main J6 SPI (OSD)	External SPI port, meant for an on-screen display board	JST-GH 1x8 (BM08B-GHS-TBT), top entry	
Main J7 AUX	Spare serial port with handshake lines	JST-GH 1x6 (BM06B-GHS-TBT), top entry	
Main J8 TELEM1	Telemetry serial port 1 with flow control	JST-GH 1x6 (BM06B-GHS-TBT), top entry	
Main J9 TELEM2	Telemetry serial port 2 with flow control	JST-GH 1x6, top entry	
Main J10 I2C	External I2C bus	JST-GH 1x4, top entry	
Main J11 RC	Serial radio control receiver port	JST-GH 1x4, top entry	
Main J12 PWM_B	Motor outputs 5 to 8 (ESC header)	JST-SH 1x8	
Main J13 PWM_A	Motor outputs 1 to 4 (ESC header)	JST-SH 1x8	

Power and debug

Main J2 POWER

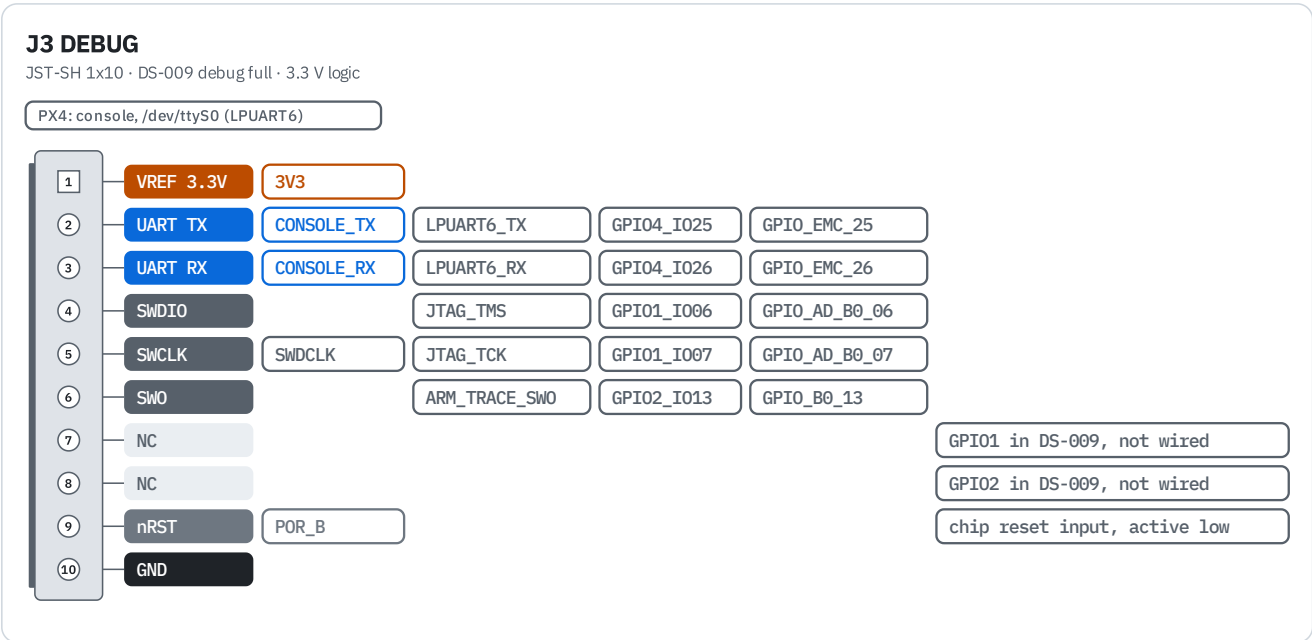


This is the main supply of the board. The power module feeds 5 V into pins 1 and 2, and the board makes its own 3.3 V and 1.8 V from it. Pins 3 and 4 are an I2C bus that reads voltage and current from the power module.

Warning

Feed only 5 V into pins 1 and 2. A higher voltage damages the board, because there is no regulator behind this connector.

Main J3 DEBUG



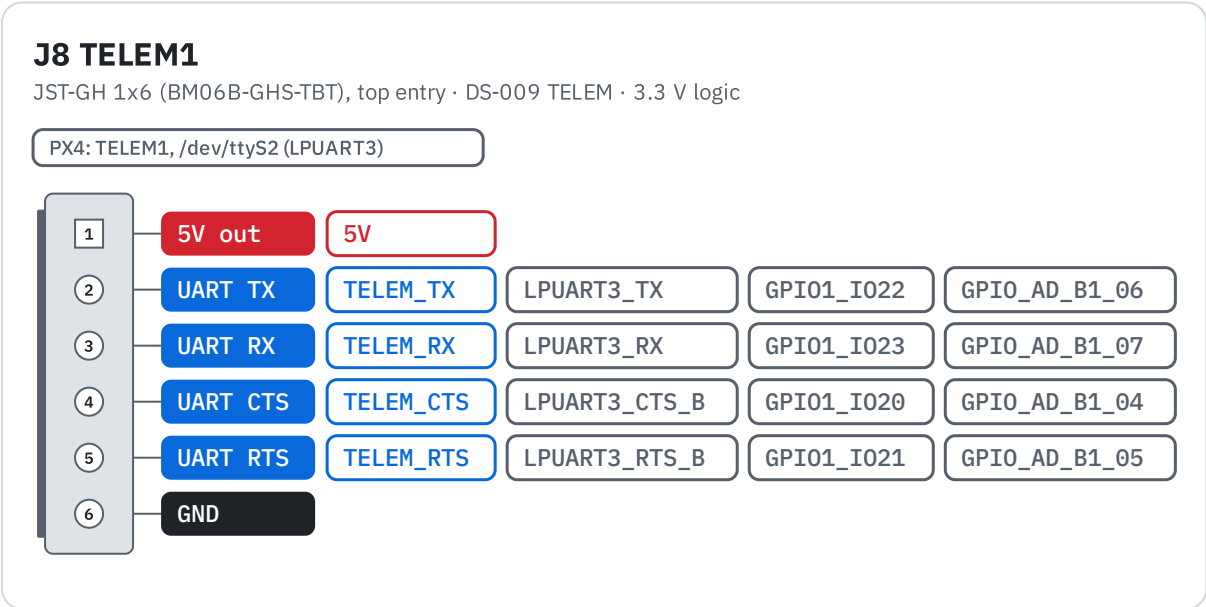
Pin 1 is a 3.3 V reference output for the debug probe, fed. Pins 2 and 3 are the serial console. Pins 4, 5 and 6 are the SWD lines of the i.MX RT1064. Pin 9 is the reset line of the chip, which is also pulled up on the board.

Warning

Pin 1 is an output at 3.3 V, not a supply input. Use a debug probe that works at 3.3 V and never drive this pin.

Serial ports

Main J8 TELEM1



Standard DS-009 telemetry layout: 5 V, TX, RX, CTS, RTS, ground. Pin 1 gives 5 V out through ferrite FL3.

Warning

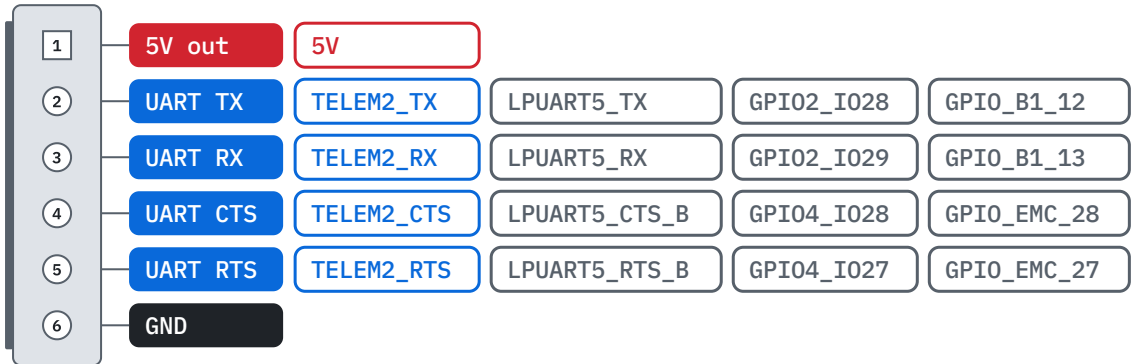
Pin 1 is a 5 V output. Never feed power into it.

Main J9 TELEM2

J9 TELEM2

JST-GH 1x6, top entry · DS-009 TELEM · 3.3 V logic

PX4: TELEM2 (LPUART5)



Same layout and same circuit as TELEM1, on a second serial port of the chip. Pin 1 gives 5 V out.

Warning

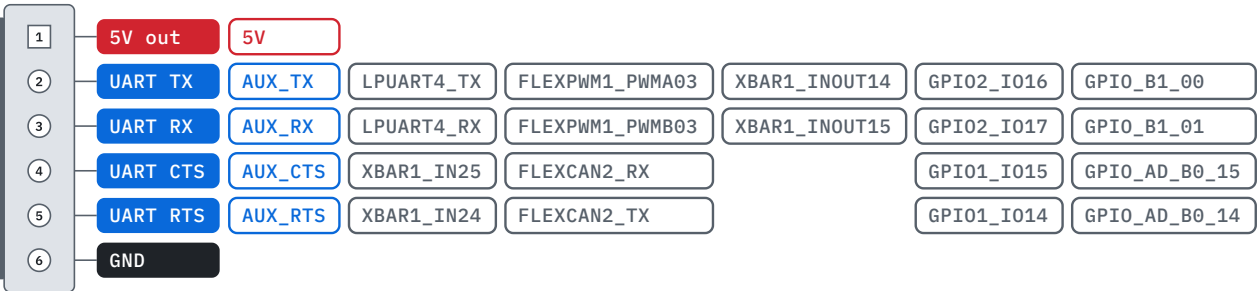
Pin 1 is a 5 V output. Never feed power into it.

Main J7 AUX

J7 AUX

JST-GH 1x6 (BM06B-GHS-TBT), top entry · 3.3 V logic

PX4: AUX, /dev/ttyS3 (LPUART4)



Pin 1 gives 5 V out. Pins 2 and 3 are LPUART4, the AUX serial port (/dev/ttyS3 in PX4). Pins 4 and 5 are called CTS and RTS in the schematic but are free pins. The chips on this connector can do more than a serial port, see the pad functions on each pin: PWM on pins 2 and 3 (FlexPWM1 module 3, A and B), a quadrature encoder or a trigger through the crossbar (XBAR1) on all four signal pins, and a second CAN bus on pins 4 and 5 (FlexCAN2, needs an external transceiver).

Warning

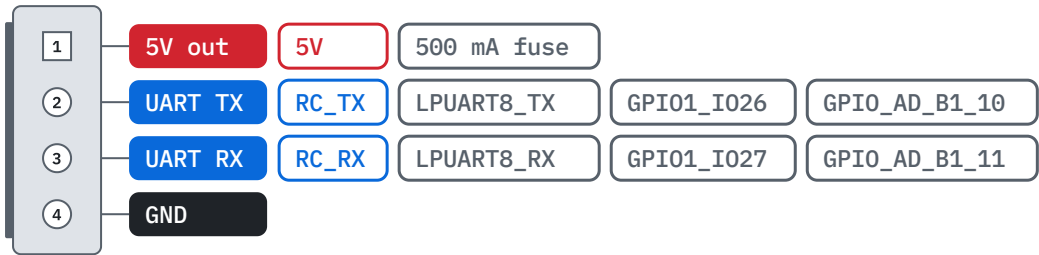
Pin 1 is a 5 V output. Never feed power into it.

Main J11 RC

J11 RC

JST-GH 1x4, top entry · 3.3 V logic

PX4: RC, /dev/ttyS6 (LPUART8)



Serial port for a digital receiver such as ELRS or CRSF. Pin 1 gives 5 V out through a resettable fuse, so a shorted receiver cannot pull down the board supply.

Warning

Pin 1 is a 5 V output. Never feed power into it.

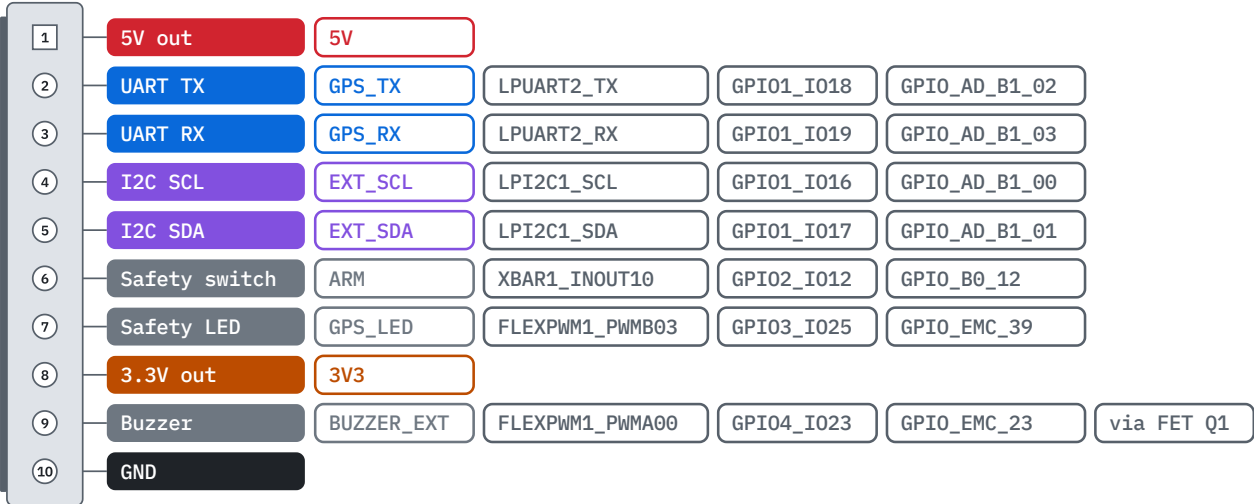
GPS, I2C and CAN

Main J4 GPS

J4 GPS

JST-GH 1x10 (BM10B-GHS-TBT), top entry · DS-009 GPS · 3.3 V logic

PX4: GPS, /dev/ttyS1 (LPUART2)



Pin 1 gives 5 V to the GPS module and pin 8 gives 3.3 V. Both are outputs. Pins 2 to 5 are the GPS serial port and the shared I2C bus, the same bus as on the I2C connector J10. Pin 6 reads the safety switch and pin 7 drives its LED. Pin 9 is the buzzer, switched to ground by FET Q1, so the board only pulls this pin low.

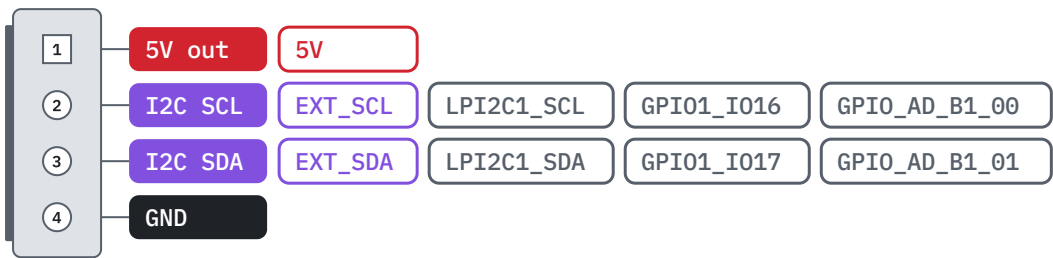
Warning

Pins 1 and 8 are power outputs. Never feed power into them.

Main J10 I2C

J10 I2C

JST-GH 1x4, top entry · DS-009 I2C · 3.3 V logic



The same bus as pins 4 and 5 of the GPS connector, so devices here share those addresses. Pin 1 gives 5 V out.

Warning

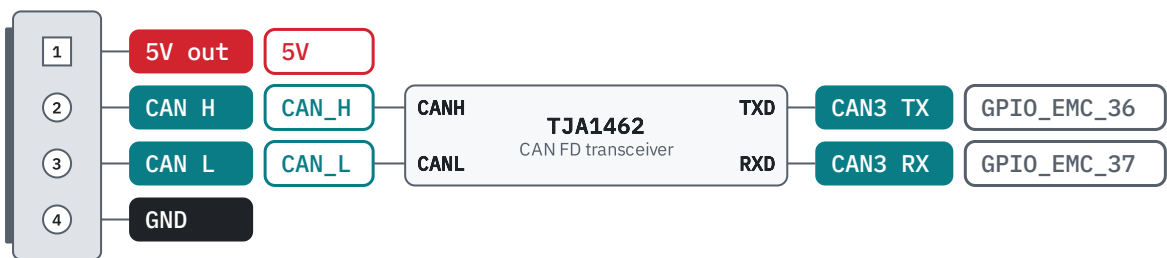
Pin 1 is a 5 V output. Never feed power into it.

Main J5 CAN

J5 CAN

JST-GH 1x4, top entry · DS-009 CAN

120 Ω termination: fitted



The TJA1462 transceiver runs from 5 V and talks to the chip at 3.3 V. CANH and CANL together form the differential bus, TXD and RXD are the logic lines to the i.MX RT1064. The 120 Ω termination is fitted, so this port is one end of the bus.

Warning

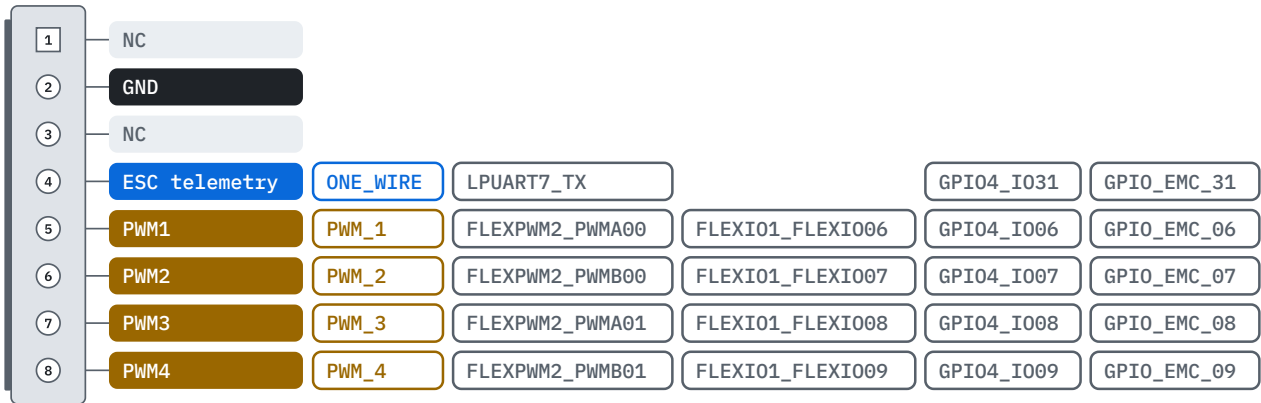
Pin 1 is a 5 V output. Never feed power into it.

Motors

Main J13 PWM_A

J13 PWM_A

JST-SH 1x8 · 3.3 V logic



Pin 4: LPUART7 in single-wire mode, receive only

Four motor signals for a four-in-one ESC. They carry normal PWM or DShot. The same pads are also FlexIO1 pins, so DShot or other protocols can be bit-banded by FlexIO. Pin 4 is the ESC telemetry line, shared with the other ESC header. The ESCs send their UART telemetry (voltage, current, RPM, temperature) on this one wire, and the chip receives it on LPUART7 in single-wire mode. It is receive only; the chip never transmits here. Pins 1 and 3 are not connected on rev F.

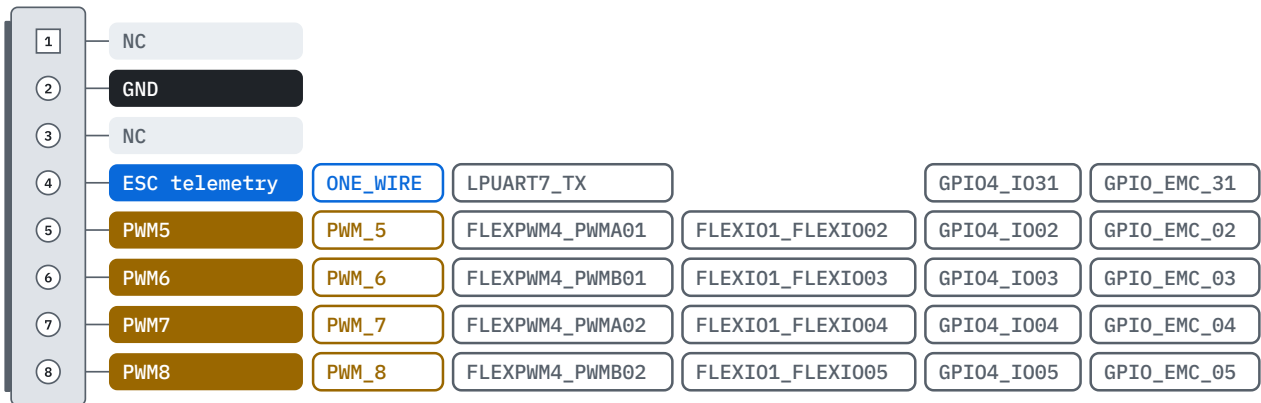
Note

This header has no 5 V pin. Many ESCs have a BEC that powers the flight controller, but this board does not use it. It gets its 5 V from the power module on J2 POWER.

Main J12 PWM_B

J12 PWM_B

JST-SH 1x8 · 3.3 V logic



Pin 4: LPUART7 in single-wire mode, receive only

Same circuit as PWM_A, for motors 5 to 8. The signals carry normal PWM or DShot. The same pads are also FlexIO1 pins, so DShot or other protocols can be bit-banded by FlexIO. Pin 4 is the same shared ESC telemetry line: LPUART7 in single-wire mode, receive only. Pins 1 and 3 are not connected on rev F.

Note

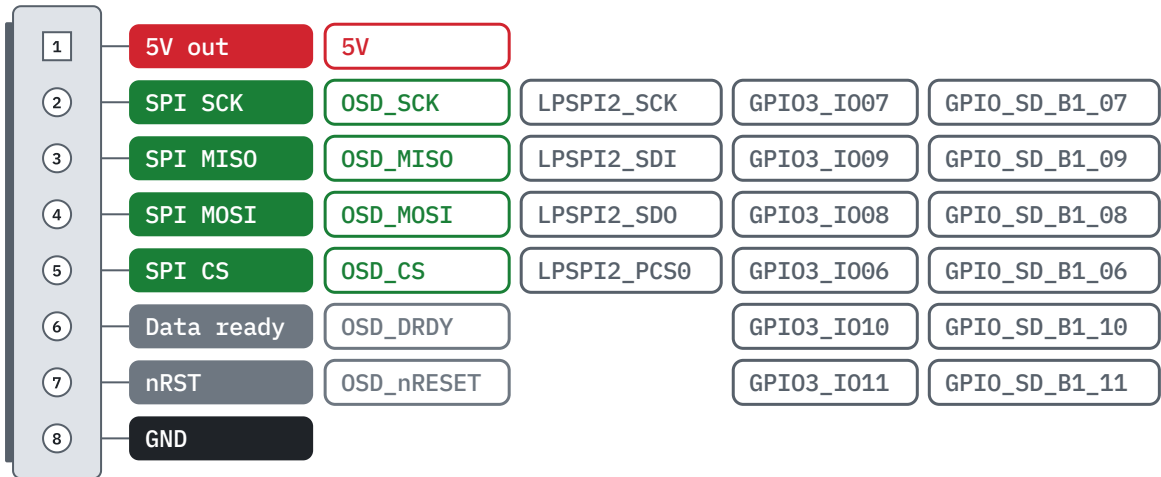
This header has no 5 V pin. Many ESCs have a BEC that powers the flight controller, but this board does not use it. It gets its 5 V from the power module on J2 POWER.

SPI and Ethernet

Main J6 SPI (OSD)

J6 SPI (OSD)

JST-GH 1x8 (BM08B-GHS-TBT), top entry · 3.3 V logic



Pin 1 gives 5 V to the add-on board, the signal lines work at 3.3 V. Pin 6 is a data-ready input and pin 7 a reset output. This port is separate from the SPI bus of the on-board sensors.

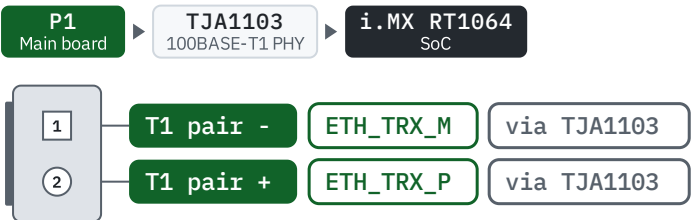
Warning

Pin 1 is a 5 V output. Never feed power into it.

Main P1 ETH

P1 ETH

JST-GH 1x2, top entry



Two wires carry send and receive at 100 Mbit/s. The TJA1103 does the physical layer and talks to the chip over RMII, so there is no direct link between these pins and a chip pad. Use a twisted pair and keep both wires the same length. The other end needs a 100BASE-T1 device or a media converter.

Main other connectors

Ref	Function	Connector	Notes
J1 USB	USB 2.0 Type-C device port	USB Type-C receptacle (USB4110-GF-A)	Serial download port of the i.MX RT1064 boot ROM, and the port used for flashing. The board is always a USB device. VBUS can also power the board through protection switch U13.
CN1 MICROSD	microSD card slot	microSD push-push, 8 pins plus card detect	Four-bit SD interface on the first SD host of the chip, used for logging. The card detect switch goes to a separate chip pin.

Buttons and switches

Ref	Board	Function	Type	Notes
SW2 BOOT	Main	Boot mode button	Push button (NANOT 240 AS)	Holding this button while the board powers up pulls boot mode 0 high and starts the serial downloader over the USB port. The second boot mode pin is tied low on the board.

Notes

- The connectors marked DS-009 on their card (TELEM1, TELEM2, GPS, I2C, CAN and the debug port) follow the Dronecode [DS-009 connector standard](#), so standard cables fit.
- The board runs from 5 V on the power module connector J2, or from USB. Every other 5 V pin is an output.

Warning

Never feed power into the 5 V pin of a JST-GH connector. Power the board through J2 or USB only.

Downloads

This reference as an [A4 PDF](#), and all cards on one A4 landscape sheet: [cheat sheet PDF](#).

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